

Analysis First-Year Exam, January 2014, Part 2

Answer *FOUR* questions. Write in a neat and logical fashion, giving complete reasons for all steps.

1. Let f be a continuous real-valued function on $[0, 1]$. For each $n \geq 1$, let $h_n = n\mathbf{1}_{[0, 1/n]}$ be n times the indicator function of $[0, 1/n]$. Prove that

$$\lim_{n \rightarrow \infty} \int_0^1 f(t) h_n(t) dt = f(0).$$

2. Let f be a continuous real-valued function on $[0, 1]$. For $n \geq 1$ and $0 \leq t \leq 1$, let $f_n(t) = f(t)t^n$. Determine a condition on $f(1)$ that is necessary and sufficient for the sequence $(f_n)_{n=1}^\infty$ to be uniformly convergent; be sure to prove necessity and sufficiency.

3. For $n \geq 1$, let f_n be a continuous real-valued function on $[0, 1]$, and for $0 \leq t \leq 1$ define

$$F_n(t) = \int_0^t f_n(u) du.$$

In each of the following cases, decide whether $(F_n)_{n=1}^\infty$ must have a uniformly convergent subsequence, giving proof or counterexample as appropriate:

- (i) $|f_n| \leq g$ for some continuous g and all n ;
- (ii) $0 \leq f_1 \leq f_2 \leq \dots$.

4. Let $(f_n)_{n=1}^\infty$ be a sequence of measurable functions on some measure space. Show that each of the following sets is measurable:

$$A = \{\omega \mid f_n(\omega) > 0 \text{ for infinitely many } n\};$$

$$B = \{\omega \mid f_n(\omega) \leq 0 \text{ for finitely many } n\}.$$

Warning: A and B need not be complementary!

5. Let $(f_n)_{n=1}^\infty$ be a uniformly-bounded sequence of continuous functions on $[0, 1]$ that converges pointwise to zero. Does it follow that

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(t) dt = 0?$$

Does the answer change if ‘uniformly-bounded’ is removed from the hypotheses? Give proof or counterexample as appropriate.